

lecture_01

August 1, 2026

1 Lecture 1: Linear Programming - Principles

{note} This lecture introduces Applied Optimization as a scientific approach to optimal decision-making and establishes the Five-Step Process as the unifying framework for the course.

Learning Objectives

By the end of this lecture, you will be able to: 1. Explain the origin and motivation for applied optimization in large, complex organisations. 2. Identify and articulate the objective, decision variables, and constraints for a given transportation problem. 3. Classify an optimization problem by its functional, temporal, and certainty properties. 4. Apply the Five-Step Process to map a real-world transportation scenario onto a structured optimization workflow.

Estimated time: 50 minutes

1.1 Origins of Applied Optimization

Human civilization witnessed a tremendous growth in size and complexity during the Industrial Revolution, leading to the segmentation of labor and management responsibilities across:

- Public organisations (e.g., national governments, port authorities, urban transport agencies)
- Private organisations (e.g., logistics firms, infrastructure developers, technology companies)

This growth gave rise to a fundamental question that motivates the entire field:

Core Question: How should available resources be allocated across operations and activities in a way that is optimal for the organisation as a whole?

Applied optimization provides a rigorous, scientific framework to answer this question — translating real-world decisions into mathematical models that can be solved systematically.

1.2 The Five-Step Process of Applied Optimization

Regardless of domain — infrastructure planning, fleet management, toll pricing, or network design — every applied optimization study follows a common five-step process. This process is the backbone of the course.

Step	Activity	Key Output
1	Define the Problem	Objective, decisions, constraints
2	Collect Relevant Data	Parameters and inputs for the model
3	Formulate the Problem	Mathematical model
4	Develop and Employ an Optimization Technique	Optimal or near-optimal solution
5	Infer and Implement Results	Actionable managerial recommendation

1.2.1 Step 1 — Define the Problem

Most problems are initially described vaguely. The first task is to precisely articulate the problem by identifying three core elements:

- **Objective** — the goal to be achieved, such as cost minimization, revenue maximization, or service level maximization.
- **Decision Variables** — the key choices that need to be made, such as how much to procure, where to locate a facility, or which route to select.
- **Constraints** — the limitations on those choices, such as budgetary limits, resource availability, regulatory requirements, or operational capacities.

Through thorough consultation, identify how all relevant stakeholders — owners, employees, suppliers, customers, regulators — influence the goal, the choices, and the limitations.

Running Example: Toll Pricing on the Chennai–Bengaluru Expressway

Consider a highway operations management firm responsible for maintaining the Chennai–Bengaluru Expressway. The firm seeks to maximize toll revenue collected from private and commercial vehicles; however, it must adhere to toll price regulations set by the National Highways Authority of India (NHAI) to ensure sufficient traffic flow and equitable access.

- Objective — maximize total toll revenue
- Decision Variables — toll price for private vehicles; toll price for commercial vehicles
- Constraints — minimum and maximum permissible toll prices for each vehicle class, as regulated by NHAI

This example will be used throughout all five steps to illustrate the complete process.

1.2.2 Step 2 — Collect Relevant Data

Once the problem’s objective, decision variables, and constraints are defined, the next task is to gather the data needed to quantify the model. This can be achieved via:

- Data management systems — for precise, real-time information (e.g., FASTag transaction logs, toll plaza sensors)
- Stakeholder consultations — for expert estimates where data is unavailable or costly to collect
- Surveys and field studies — for primary data on demand, behavior, or operational parameters

Running Example (continued)

For the toll pricing problem, relevant data includes: - FASTag data — vehicle counts, vehicle class breakdown, and transaction timestamps for tagged vehicles - Manual traffic counts and toll booth surveys** — flow estimates for untagged vehicles - NHAI tariff schedules — current regulatory bounds on permissible toll prices - Demand elasticity estimates — how traffic volume responds to changes in toll price, sourced from prior studies or consultant reports

{note} Data quality directly determines model quality. A well-formulated model built on poor data will produce unreliable recommendations. Identifying data gaps early is a critical part of this step.

1.2.3 Step 3 — Formulate the Problem

A well-defined problem with relevant data enables us to formulate a concise mathematical model — a system of equations that defines the cause-effect relationship between the objective function, decision variables, and constraints.

All models are idealised representations of reality, involving necessary approximations and assumptions. Problem formulation is therefore an iterative process of model enrichment: develop, validate, refine.

General Form A typical optimization model with objective function f , decision variable vector $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, and constraint vector $\mathbf{g}$ takes the following standard form:

$$\max_{\mathbf{x}} f(\mathbf{x})$$

Subject to:

$$\mathbf{g}(\mathbf{x}) \leq \mathbf{0}$$

{note} **Constraint conversions to standard form:** A $\geq$ constraint can be converted as: $\mathbf{g}(\mathbf{x}) \geq 0 \implies -\mathbf{g}(\mathbf{x}) \leq 0$. An equality constraint requires two inequalities: $\mathbf{g}(\mathbf{x}) = 0 \implies \mathbf{g}(\mathbf{x}) \leq 0$ and $-\mathbf{g}(\mathbf{x}) \leq 0$.

Problem Classification Optimization problems — and the models used to represent them — can be classified along three dimensions:

Dimension	Type A	Type B
Functional	Linear Problem	Non-Linear Problem
Temporal	Static Problem	Dynamic Problem
Certainty	Deterministic Problem	Stochastic Problem

This course focuses on linear and non-linear formulations under static, deterministic settings. Dynamic and stochastic extensions are important but lie beyond our scope.

We will explore specific formulations — the transportation problem, assignment problem, TSP, VRP, and others — from Lecture 2 onwards.

1.2.4 Step 4 — Develop and Employ an Optimization Technique

Once the problem is formulated, the goal is to find values of the decision variables that optimize the objective function while satisfying all constraints. Two broad families of methods exist:

Approach	Description	Example Methods	Best For
Exact methods	Guarantee the mathematically optimal solution	Simplex algorithm, interior-point methods, branch-and-bound	Small to medium problems with well-structured formulations
Non-exact methods	Find high-quality, near-optimal solutions efficiently	Local search heuristics, evolutionary algorithms, swarm intelligence	Large-scale or computationally intractable problems

The choice of method depends on: - The size and complexity of the problem - The structure of the objective function and constraints (linear vs. non-linear) - The available computational resources and time budget - The required solution quality (provably optimal vs. practically good)

Running Example (continued)

The toll pricing problem — maximizing revenue subject to regulatory bounds — is a linear program (if demand is treated as fixed) or a non-linear program (if demand responds elastically to price). For the linear case, the simplex algorithm yields the exact optimal toll prices. For the non-linear case, methods such as penalty, barrier, or interior-point methods would be appropriate.

`{important}` A solution is optimal or near-optimal only ****with respect to the model****, not necessarily the real-world problem. If the model is appropriately validated, the solution should be a good approximation for practice - but this cannot be assumed without validation.

1.2.5 Step 5 — Infer and Implement Results

Finding an optimal solution is not the end of the process — it is the beginning of decision-making. In real-world applications, no single solution should be implemented without scrutiny. **Post-optimality analysis** is the set of techniques used to interpret, stress-test, and contextualise the solution before acting on it.

Post-optimality analysis includes:

- **Sensitivity analysis** — determines which model parameters most critically influence the optimal solution, and quantifies how the solution changes as those parameters vary within a range. This is essential for understanding robustness.
- **What-if analysis** — evaluates how the optimal solution changes under different structural assumptions (e.g., different demand scenarios, regulatory changes, or new constraints).
- **Managerial interpretation** — translates the mathematical solution into actionable recommendations that account for practical, institutional, and stakeholder considerations not captured in the model.

Running Example (continued)

Having solved the toll pricing model, the firm would perform sensitivity analysis to understand: - How much does optimal revenue change if NHAI tightens the maximum permissible toll by 5? - What is the revenue impact if commercial vehicle traffic falls by 10% due to an alternative route opening? - Is the optimal solution robust across different demand elasticity estimates?

These questions — not the raw optimal values — are what drive the managerial recommendation.

`{important} **Objective 3 of this course** is specifically about this step: interpreting solutions derived from optimization models and making informed managerial decisions in real-world transportation scenarios. Post-optimality analysis will be revisited at the end of every major module.`

1.3 Take-Away Exercise

Work through the Five-Step Process for the following scenario. You do **not** need to solve the mathematical model — focus on Steps 1 and 3 (problem definition and classification).

Scenario: Last-Mile Delivery at a Chennai Fulfilment Centre

A Flipkart fulfilment centre in Chennai operates a fleet of electric delivery vehicles. Each morning, the operations manager must assign parcels to vehicles and plan delivery routes for the day. The centre has 5 vehicles, each with a maximum payload of 200 kg. There are 40 customer delivery points spread across the city. Each delivery point has a known demand (weight of parcels) and a preferred delivery time window. The objective is to serve all customers at minimum total travel distance, while respecting vehicle capacity and time window constraints.

Your tasks:

1. Identify the objective, decision variables, and constraints. Who are the relevant stakeholders, and how do they influence each element?
2. Classify the problem along the three dimensions (functional, temporal, certainty). Justify each classification.
3. Identify one key data source you would use for each of the three problem elements (objective, decisions, constraints).

4. (*Stretch*) Would you use an exact or non-exact solution method for this problem? Justify your answer in one sentence.
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{important} This course focuses on Steps 1, 3, 4, and 5 - defining and formulating transportation problems, developing and employing appropriate optimization techniques, and inferring and implementing results. Step 2 (data collection) is addressed in the context of specific applications.

1.4 Further Reading

- Hillier, F.S. and Lieberman, G.J. (2015) — *Introduction to Operations Research*, 10th ed. — McGraw-Hill (*Chapters 1–2: Introduction and Overview of OR*)
- Taha, H.A. (2017) — *Operations Research: An Introduction*, 10th ed. — Pearson (*Chapter 1: What is OR?*)
- Bazaraa, M.S., Sherali, H.D., and Shetty, C.M. (2006) — *Nonlinear Programming: Theory and Algorithms*, 3rd ed. — Wiley (*Chapter 1: Introduction*)